

INF2003: Database Systems

Lab 01: Python-based Database

Objectives

To learn the basic database operations via Python and SQLAlchemy.

Background

We have not yet introduced the technical details of a database system in the first week, e.g., SQL coding. In this week's lab, we aim to understand the basic database operations, based on our existing (hopefully) skillset, Python.

We will use Object-Relational Mapping (ORM), it enables developers to interact with a database using the concepts and syntax of their chosen programming language's object model, rather than writing raw SQL queries. Specifically, there is a Python library, called SQLAlchemy, which allows us to bypass the SQL statements and perform the database operations in Python. We will learn how to create a database and a table with SQLAlchemy first. Accordingly, we perform the basic CRUD (create, read, update, and delete) operations. After the lab, you would be well prepared for the next week's lecture on SQL.

Software and References

- We will use Google Colab. A brief introduction on how to use Google Colab can be found [here](#).
If you have Jupyter notebook installed on your local machine, you can also work locally. Both Jupyter and Colab should work similarly. A free version of Colab account should be sufficient, you are not required to subscribe to pro account.
- If you are not familiar with Python programming, there are plenty of resources that can be found online. For example:
<https://python.land/python-tutorial>
- A tutorial about SQLAlchemy can be found at: <https://www.sqlalchemy.org/>
Please read the tutorial and understand the syntax and functions.

Instructions and Deliverables

1. You may download the 'INF2003_Lab01_student.ipynb' from Xsite and open it in Jupyter notebook or Google Colab.
2. You are required to finish the questions listed in the 'INF2003_Lab01_student.ipynb' file and insert your code in cells after the questions. When you have completed all the questions, download your .ipynb file, rename it as: <lab id>_<student id>_<name>_Lab1.ipynb, then submit it to the Xsite Dropbox folder under: lab1.

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Note: please keep the instructions in the .ipynb file, only insert your code in the designated cells. Please run your code to check the output when submitting, make sure you have kept the execution output.

Format: **ONE (1) IPYNB file**

Filename: <lab group id>_<student id>_<name>_Lab1.ipynb.

Venue: Dropbox in xSiTe.

Deadline: The following **Monday 11:30pm** when the lab is conducted. Extension requests, if any, shall be submitted to the lecture before the deadline, and the applications will be reviewed on a case-by-case basis.

Penalty: A penalty of 100% for this lab will be imposed for the 1st time plagiarism and a penalty of 100% for ALL the labs for the 2nd time plagiarism.